



Research Paper

Adaptively multi-modal contrastive fusion network for molecular properties prediction

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ARTICLE INFO

Keywords:

Multi-modal fusion

Prior knowledge

Multi-head attention mechanism

Contrastive learning

Molecular property prediction

ABSTRACT

Molecular property prediction has become the mainstream approach for revealing the underlying mechanisms of biomedical systems with molecular representations. Existing prediction methods based on deep learning typically learn features from molecules at a specific modality or simple fusion solution, failing to consider the inconsistency, complexity, and relationships inherent in multi-modal data. To solve this issue, an adaptively multi-modal contrastive fusion network (AMCFNet) is proposed to adaptively extract the complementary features from interaction and consensus between multi-modal representations for molecular property prediction of breast cancer. The proposed model begins with a two-stream feature extractor module, which learns both one-dimensional (1D) and two-dimensional (2D) molecular representations simultaneously. The basic part of the network is the adaptively contrastive fusion module, contrastively learning features between similar and different molecules with consensus scores, which can adaptively allocate weight to fuse semantic and structural information while avoiding cognitive gaps caused by inconsistencies within multi-modal. Additionally, the final complementary molecular representation is derived by integrating 1D, 2D, and fused 1D-2D features to enhance the prediction of molecular properties in breast cancer. The proposed AMCFNet model is evaluated on five estrogen receptor alpha (ER α) and five compound public datasets, consistently outperforming state-of-the-art baselines in classification and regression tasks of molecular property prediction including single- and multi-modal methodologies.

1. Introduction

Molecular property prediction is a pivotal task for the synthesis of chemical structure and biological characteristics. Through the analysis of properties such as absorption, distribution, metabolism, extraction, and toxicity (ADMET) of drug molecules, researchers are able to gain insights into the chemical properties and drug interactions of candidate drugs. In breast cancer treatment, accurate prediction of ADMET properties is very important, which enables the effective solution of drug-target interaction and drug discovery while minimizing risks of failure. Molecules can be represented as descriptors, fingerprints, SMILESs (Simplified Molecular Input Line Entry Systems), graphs, and so on. Methods based on the integration of machine learning (ML) and deep learning (DL) are widely employed for tasks like drug discovery (Li et al., 2021; He et al., 2023) and medical diagnosis (Anari et al., 2022; Ranjbarzadeh et al., 2023).

In recent years, diverse methods have been designed for predicting molecular properties, such as SMILES-based Transformer (Ross

et al., 2022; Monteiro et al., 2023; Honda et al., 2019), SMILES-based BERT (Liu et al., 2023; Zhang et al., 2022), GCN (Graph Convolutional Networks) (Duvenaud et al., 2015), and GAT (Graph Attention Networks) (Velickovic et al., 2017), etc. These models are deep neural networks that extract features from specific molecular representation. For example, Shi et al. (2023) introduced the ensemble learning algorithm to predict the molecular ADMET properties by descriptor data. Ding et al. (2022) combined molecular multi-fingerprints with machine learning models to predict the hERG cardiotoxicity of compounds. Chen et al. (2023b) employed the pyramid network structure and molecular SMILES representation to predict drug-target binding affinity. Chen et al. (2023a) partitioned molecular graph data into multiple functional group-based clusters for discovering structure–property relationships. Extensive work towards molecular feature extraction targets specific representation analysis but ignores the synergistic and complementary molecular information available in diverse modalities. This limitation results in challenges in molecular representation learning and reduces

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model accuracy. Thus, it is important to develop reliable approaches to combine different molecular representations for learning a more comprehensive range of molecular features.

As new technologies advance in drug discovery, some multi-modal methods have been introduced to provide valuable insights into the mechanisms of molecular properties and structures. For example, Liu et al. (2021) utilized 2D topological structures and 3D geometric perspectives for self-supervised learning and capturing high-level features in molecule data. Zhang et al. (2024) utilized 1D, 2D, and 3D molecular representations with distinct pre-trained methods to predict molecular properties. Wu et al. (2023) performed molecular joint representation learning via molecular SMILES and graphs. These methods can efficiently extract molecular features from different modal data using various techniques. While the methods yield promising outcomes with the combined techniques considered. All of the above multi-modal fusion analyses of molecular properties almost conduct a simple integrating strategy. The inconsistencies, complexities, and interrelations inherent in diverse representations make it challenging for the model to effectively focus on the integration of molecular complementary information. Current works largely overlook the crucial interactions among molecular multi-modal data, resulting in certain errors in cognitive gaps.

Therefore, in practical scenarios, some challenges affect the molecular property prediction work. (1) Multi-modal fusion challenge. Most of the existing models only consider rudimentary fusion between different modalities, ignoring the adaptive interaction fusion of molecular features. (2) Prediction accuracy challenge. The in-depth analysis of molecular property remains inadequate, failing to fully leverage and combine the advantages of various single-representation methods. The model's generalization and learning capability may be constrained, leading to a decline in performance.

In light of these challenges, we develop an adaptively multi-modal contrastive fusion network (AMCFNet) by integrating the SMILES and graph representations for molecular ADMET properties prediction. Our framework consists of three modules: the prior K-BERT module, developing for extracting 1D-SMILES molecular information. The multi-layer GNN module is responsible for learning the molecular 2D structure of the graph representation. And the adaptively contrastive fusion module, which is designed to adaptively extract the complementary features by contrastively and interactively learning on different and similar molecules via a consensus function. This way, we can integrate all molecular modalities by concatenating the 1D-SMILES, 2D-graph, and fused SMILES-graph to an FC layer to predict molecular ADMET properties. In summary, the AMCFNet model enables a comprehensive understanding of the relationship between the semantic information and structural properties of molecules, improving the performance and the fusion ability of the model. From the application's perspective, it can accurately predict the molecular ADMET properties under multi-modal analysis, accelerating the process of bioinformatics research and drug discovery. Extensive experiments conducted on benchmarks demonstrate the effectiveness and generalization ability of our AMCFNet method in fusing the multi-modal data and accurately classifying the molecular ADMET properties' utility.

Our contributions can be summarized as follows:

- (1) We propose an adaptively multi-modal contrastive fusion deep learning neural network (AMCFNet) for predicting molecular ADMET properties in breast cancer. This network can comprehensively and adaptively extract both semantic information and structural properties of molecular features, enabling accurate prediction of molecular properties.
- (2) In AMCFNet, the prior K-BERT and Multi-layer GNN modules are designed for respectively extracting detailed 1D-SMILES and 2D-graph molecular features, aiming to avoid misunderstanding caused by inaccurate or incomplete chemical characterizations extraction.

- (3) We construct an adaptively contrastive fusion module to contrastively learn molecular features with consensus scores, which can adaptively fuse the semantic and structural information from SMILESs and graphs, considering both inconsistencies and similarities between modalities.
- (4) Extensive experiments demonstrate that the proposed AMCFNet framework outperforms state-of-the-art methods significantly on the breast *Erα* datasets and five publicly available datasets, and promises to integrate molecular data.

The rest of the paper is organized as follows: In Section 2, we refer to the related work associated with this paper. In Section 3, we elaborate on the details of our proposed AMCFNet framework. In Section 4, we demonstrate the experimental results of several real-world datasets and the analysis with the proposed AMCFNet framework. In Section 5, we contain the conclusion and suggested future work.

2. Related works

2.1. Single-modal molecular representation learning

Molecules can be represented as descriptors, fingerprints, SMILESs, graphs, and so on. Each modality provides unique information about the target, revealing diverse characteristics. Recently, approaches combining ML and DL have been extensively utilized for tasks such as feature extraction and molecular property prediction, including fingerprints-based methods (Ding et al., 2022; Riniker, 2017; Rogers and Hahn, 2010), descriptor-based methods (Shi et al., 2023; Rahul and Clement, 2023; Comesana et al., 2022), SMILES-based methods (Chen et al., 2023b; Kimber et al., 2021; Sharma et al., 2024), and graph-based methods (Chen et al., 2023a; Qu et al., 2024; Zhang et al., 2023).

For instance, Ding et al. (2022) combined the MDFFP, Baseline2D, ECFP4, and PropertyFP of molecular fingerprints with machine learning models to predict the hERG cardiotoxicity of compounds. However, molecular fingerprints, represented as fixed-length binary vectors, may lose some detailed structural information. For molecular descriptor representation, the QSAR (Quantitative Structure-Activity Relationship) (Song et al., 2024; Wu et al., 2024) and ML methods (Zhao et al., 2023; Ji et al., 2023) are used to establish mathematical relationships between molecular structures and their biological activities. Shi et al. (2023) introduced the ensemble learning algorithm to analyze important features by molecular descriptor data and predict the molecular ADMET property. Yet, descriptors are typically numerical features of molecular properties that require manual selection, potentially failing to capture the complexity of molecules comprehensively. Additionally, Chen et al. (2023b) employed the pyramid network structure and molecular SMILES representation to fuse features extracted from multi-scale layers to predict drug-target binding affinity. And owing to its succinct textual representation, natural language processing models find extensive application in molecular property prediction, such as SMILES-based Transformer (Ross et al., 2022; Monteiro et al., 2023; Honda et al., 2019), and SMILES-based BERT (Liu et al., 2023; Zhang et al., 2022). However, SMILES lacks direct numerical information about the molecule, which may not be explicit enough for certain structural information. Furthermore, Chen et al. (2023a) partitioned molecular graph data into multiple functional group-based clusters and designed the Segmentation MPNN to learn function groups for discovering structure-property relationships. While many researchers utilize 2D-graph structures to extract molecular features like GNN, GCN (Duvenaud et al., 2015), GAT (Velickovic et al., 2017), and GIN (Xu et al., 2018). However, graph models deal with more complex structural information and may require more significant computational resources.

Although these methods can effectively extract the features from each molecular modality. None of the above modalities can comprehensively analyze the molecular characteristics, resulting in an inability to

harness the strengths of multi-view information. Consequently, molecular feature integration based on multi-modal garners more attention due to its enhanced performance compared to single-modal methods.

2.2. Multi-modal molecular representation learning

Multi-modal learning techniques are currently being applied across various domains, with particular significance in bioinformatics and drug discovery, as a pivotal approach for analyzing the properties and interactions of biological molecules. For example, Liu et al. (2021) proposed a GraphMVP framework, designed to leverage the correspondence and consistency between 2D topological and 3D geometric structures for self-supervised learning and capturing high-level features in molecule data. However, the 3D geometry knowledge was pre-trained with a 2D feature encoder and later fine-tuned for downstream tasks that excluded 3D perspectives, potentially overlooking complementary features during interactions. Zhang et al. (2024) designed a PremuNet network, which utilizes 1D, 2D, and 3D molecular representations with distinct pre-trained methods to predict molecular properties and employs GNN to integrate multi-modal molecular features. Yet, the method only adopted a simple concatenate strategy for fusion, failing to adequately account for the interaction and relationships of the multi-modal representation. Wu et al. (2023) proposed an MMSG framework to perform molecular joint representation learning via molecular SMILESs and graphs while introducing BMC-GNN to enhance the aggregation of graph information and facilitate further combination. Nevertheless, the model employed the graph information to directly update the SMILES representation for message communication and integration, which may neglect potential inconsistencies between the multi-modal features.

Although these networks achieve good performance in extracting and integrating three-dimensional molecular features, they mainly suffer from some limitations. They only consider a basic fusion method for multi-modal data, ignoring the differences in contribution of each representation to the model. Additionally, the inconsistency, complexity, and relationships inherent in multi-representation data are crucial for analysis in multi-modal scenarios. However, these methods fail to support the contrastive and interactive learning for the mentioned correlations among different modalities, resulting in a relatively narrow understanding of features.

In this work, we propose an adaptively multi-modal contrastive fusion network (AMCFNet), whose goal is to jointly understand molecular information in various modalities for molecular ADMET properties prediction. Initially, we develop a SMILES-based feature extraction module and a graph-based feature extraction module to learn molecular features from SMILESs and graphs, respectively. Then, we construct an adaptively contrastive fusion module to integrate SMILES and graph features, creating a complementary molecular representation. In essence, we emphasize the importance of contrastive and fusion relationships, ultimately improving the model's accuracy in predicting complex molecular ADMET properties. Additionally, our AMCFNet model is based on the correlation, contribution, and adaptive fusion between multi-modal data, enabling it to learn richer interactive and comprehensive information across different levels of each modality characteristic.

3. Proposed method

AMCFNet comprises three modules: prior K-BERT module, multi-layer GNN module, and adaptively contrastive fusion module. The flowchart of AMCFNet is shown in Fig. 1(A). Specifically, the prior K-BERT module, developing to extract 1D-SMILES global molecule-level features from SMILES strings. The multi-layer GNN module is employed to extract graph structure-level molecular features from the molecular graphs. Subsequently, we construct an adaptively contrastive fusion

module to perform the adaptive weight allocation for features integrating with contrastive learning strategy. After that, we concatenate the molecular 1D-SMILES, 2D-graph, and fused SMILES-graph features for downstream ADMET properties prediction tasks. With this method, we can adaptively fuse the 1D-semantic and 2D-structure molecular features and comprehensively understand the multi-modal molecular representations to accurately predict molecular ADMET properties.

3.1. Prior K-BERT module for SMILES feature extraction

Molecular semantic information learning has important applications in drug discovery. Wu et al. (2022) developed a pre-training method known as K-BERT to extract chemical information from SMILES by employing three pre-training tasks: atomic feature prediction, molecular feature prediction, and contrastive learning. These tasks collectively enhance the model's robustness against small disturbances, such as synonyms. We introduce the K-BERT network based on prior knowledge for extracting global molecule-level features from 1D-SMILES representations. The flow chart is illustrated in Fig. 1(A)(I). Six transformer coding layers are designed for K-BERT to extract the features and relationships in the molecular SMILES. Assuming that $Input_S \in R^{N \times 1}$ is the input SMILES representation, N is the number of molecules, our embedding mechanism for this SMILES modality is:

$$X_{eS} = \text{LN}(\text{tok}_e(\text{Input}_S) + \text{pos}_e(\text{pos})), \text{pos} \in (0, l_S) \quad (1)$$

where $X_{eS} \in R^{N \times l_S}$ is the embedding vector by embedding mechanism; l_S is the length of SMILES token sequence; $\text{pos} \in R^{N \times l_S}$ is the position information in the SMILES token sequence; tok_e is the token embedding layer; pos_e is the position embedding layer; LN is the Layer Normalization.

SMILES features are directly extracted by the K-BERT method may lose domain-specific information. To learn more task-relevant information, we employ the parameter fine-tuning strategy in the current molecular ER α dataset of breast cancer and define the training process as follows:

$$X_S = \text{FC}(\text{K-BERT}(X_{eS})) \quad (2)$$

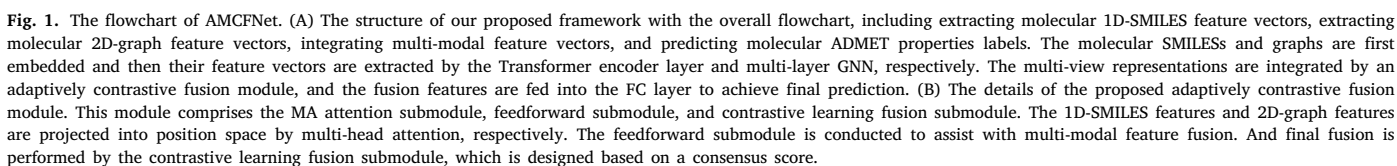
$$\text{K-BERT} = I_{Tf}(\text{F}_{Tf} \circ 5) \quad (3)$$

where $X_S \in R^{768 \times 1}$ is the extracted SMILES features; FC is the fully connected layer; F_{Tf} is the freezing of the transformer encoder layer; I_{Tf} is the initialization of the transformer encoder layer; $\circ$ is the operation operator. Specifically, the parameters of the first 5 transformer layers remain frozen, preserving knowledge from the prior task. Then, the final Transformer layer is trained from scratch, optimizing parameters based on the new dataset.

By introducing the K-BERT method, the 1D-SMILES semantic features can be extracted and alleviate the model's susceptibility to small disturbances such as synonyms. With the advantage that after parameter fine-tuning, the model can reduce the training time and resource consumption while maintaining high performance.

3.2. Multi-layer GNN module for graph feature extraction

The molecular graph that covers both local and global characteristics of molecules could be significantly versatile and flexible. It is necessary to capture the global information among nodes while dynamically adjusting the degree of attention between nodes, which is why integrating GCN and GAT is critical. Here we propose an efficient approach to capture local and global information of diverse input nodes and channels for extracting the molecular 2D-graph structure features. Specifically, the module comprises two GCN layers, two GAT layers, four ReLU (Rectified Linear Unit) layers, and a GAP (Global Average Pooling) layer. The process of this module is shown in Fig. 1(A)(II).


$$\text{ReLU} = \max(0, X) \quad (7)$$

where H denotes the height of feature map, W denotes the width, and $x_{i,j}$ denotes the feature value at position (i, j) .

We develop the Multi-layer GNN to extract the 2D-graph features, which can effectively aggregate molecular structural features from neighboring nodes and capture both global and local features simultaneously.

3.3. Adaptively contrastive fusion module

Under the guidance of DL multi-modal fusion strategies, depth fusion technology can learn more comprehensive features that are challenging to compare with simple concatenation methods. For different modal features of the same object, the semantic distinction and modal consistency between modalities are required. Therefore, we design an adaptively contrastive fusion module for integrating 1D-SMILES semantic information and 2D-graph structural properties, which can measure the similarity between different modalities and the same modalities while avoiding information loss caused by direct concatenation. As shown in Fig. 1(B), this module comprises MA attention submodule, feedforward submodule, and contrastive learning fusion submodule.

3.3.1. MA attention submodule

Supposed we have two modalities as input: molecular 1D-SMILES features $X_S \in R^{768 \times 1}$ and 2D-graph features $X_g \in R^{64 \times 1}$, obtained by prior K-BERT module and multi-layer GNN module, respectively. We introduce two multi-head attention modules dedicated to $X_S \in R^{768 \times 1}$ and $X_g \in R^{64 \times 1}$, our projection for these two modalities is:

$$X_{SM} = MA(X_S) \quad (10)$$

$$X_{gM} = MA(X_g) \quad (11)$$

$$MA(Q, K, V) = \text{Concat}(\text{head}_1, \text{head}_2, \dots, \text{head}_h) \cdot W^O \quad (12)$$

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \quad (13)$$

where MA is the multi-head attention mechanism, which utilizes the learnable weight matrices and to project the $X_S \in R^{768 \times 1}$ and $X_g \in R^{64 \times 1}$ features to Q(Query), K(Key), and V(Value) vector spaces, respectively; $Q_i = XW_i^Q$, $K = XW_i^K$, and $V = XW_i^V$; h is the number of attention head; W^O , W_i^Q , W_i^K , and W_i^V are learnable matrices.

For this study, an 8-head attention mechanism is employed to enhance the model's capacity for capturing diverse relationships and focusing on distinct positions within the input features.

3.3.2. Feedforward submodule

The dimensions and information between different modalities are relatively complex. As a further step towards enabling better fusion, we define a feedforward submodule that takes the MA projection features $X_{SM} \in R^{768 \times 1}$ and $X_{gM} \in R^{64 \times 1}$ as input. The details are defined as follows:

$$X_{N(S-g)} = \text{LN}(\text{ReLU}(\text{FC}(X_{SM})) + \text{ReLU}(\text{FC}(X_{gM}))) \quad (14)$$

where FC is the fully connected layer, which can align their dimensions with the two modalities and effectively reduce redundancy; ReLU is the activation function; LN is the layer normalization, to address internal covariate shift issues in deep neural networks, ensuring each layer's input maintains a stable mean and variance. Following this, to approximate complex nonlinear functions, we define the training formula as:

$$X_{S-g} = \text{FC}(\text{ReLU}(\text{FC}(X_{N(S-g)}))) \quad (15)$$

where $X_{S-g} \in R^{128 \times 1}$ is the feedforward representation by two ReLU functions and one FC layer. The feedforward submodule can facilitate the model's ability to capture intricate relationships within the features.

3.3.3. Contrastive learning fusion submodule

However, the information contained between different modal features is inconsistent and complex, fusing their complementary representations is extremely challenging with just a simple concatenation. So, we design a contrastive learning submodule to the multi-modal fusion by using the consensus score. From Fig. 1(B), it can be seen that the contrastive learning submodule consists of two tasks: consensus similarity score calculating and different similarity score calculating.

The consensus similarity score is derived by measuring the similarity between the representations of the fused features across multiple modalities, emphasizing consistency and coherence between them. To be specific, we first calculate the same molecular similarity based on the canonical SMILES by cos function Eq. (16).

$$\cos(F_a, F_b) = \frac{F_a \cdot F_b}{\|F_a\| \|F_b\|} \quad (16)$$

where $\cos(\cdot)$ is a cosine similarity function; F_a and F_b are the different molecular features; $\|\cdot\|$ is the Euclidean norm function. Then, we introduce the FC layer and Sigmoid function to perform non-linear transformations and use its result as the weight value that is used as the contribution of the contrastive learning to adaptively learn the multi-modal based on similar molecules. To assess how well the fusion of features from different modalities aligns with a shared semantic space, we multiply the similarity and weight according to Eq. (17) to obtain the summed score. After that, the consensus score is evaluated by Eq. (18), which sums the four similarity values to capture the agreement between modal representations. This score acts as a guiding signal for the model to identify the most consistent features and minimize the impact of noise or inconsistencies.

$$\text{Sum}_s = \sum_{d \in D_n} (\cos(F_{n,c}, F_{n,d}) \times \text{Sigmoid}(\text{FC}(F_{n,d}))) \quad (17)$$

$$\text{Consensus}_s = \sum_{n=1}^N \frac{4 - \text{Sum}_s}{\text{Sum}_s} \quad (18)$$

where Consensus_s is the consensus similarity score; N denotes the batch size; n denotes the current molecule; D_n denotes four distinct representations generated from the standard representation for the current molecule; $F_{n,c}$ denotes the feature of standard representation for the current molecule; $F_{n,d}$ denotes the feature of four enhanced representations for the current molecule. Meanwhile, we calculate the molecular different similarity score based on every canonical representation by Eq. (16). This score contrasts how distinct each molecular fused representation is from the others and emphasizes the diversity between modal features, encouraging the model to retain the unique information from each modality.

$$\text{Different}_s = \sum_{n=1}^N \sum_{m \in B_n} \cos(F_{n,c}, F_{n,m}) \quad (19)$$

where Different_s is the different similarity score; m denotes one of the remaining standard molecules in the batch; B_n denotes the other standard molecules remaining in this batch; $F_{n,m}$ denotes the feature of the standard representation in this batch.

To this end, we can get the contrastive learning loss by adding the two consensus scores and obtain the contrastive fused multi-modal features $X_{F(S-g)} \in R^{128 \times 1}$, which are used as the SMILES-graph fused features in the next. The contrastive learning loss function is defined as below:

$$\text{Loss}_{CL} = \text{Consensus}_s + \text{Different}_s \quad (20)$$

where Loss_{CL} is the contrastive learning loss function, which maximizes the similarity among distinct features corresponding to the same molecules and minimizes the similarity between features of different molecules. Our proposed AMCFNet captures both the overall consistency between modalities and their individual characteristics through the contrastive learning loss function. To achieve this, the model uses

a sigmoid function to assign adaptive weights to the similarity scores. By adaptively learning from both shared and distinct features across different modalities, the model dynamically adjusts the contribution of each similarity score. This process assigns higher weights to more informative similarities during the training phase, while minimizing the impact of less informative ones.

Based on the $X_S \in R^{768 \times 1}$, $X_g \in R^{64 \times 1}$, $X_{F(S-g)} \in R^{128 \times 1}$, we can concatenate the features extracted from the 1D-SMILES, 2D-graph, and fused SMILES-graph representations, consolidating comprehensive and enriched information to obtain the final multi-modal fusion features $X_F \in R^{960 \times 1}$.

$$X_F = \text{Concat}(X_S, X_g, X_{F(S-g)}) \quad (21)$$

The objective of our method is to classify molecular ADMET properties with breast cancer. Therefore, the binary cross-entropy loss function is employed to optimize, which is defined as:

$$Loss_C = -\frac{1}{N} \sum_{i=1}^N [y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)] \quad (22)$$

where N is the number of molecules; $\log$ is the natural logarithm; y_i is the actual binary label for the i th molecule (0 or 1); $\hat{y}_i$ is the predicted probability that the i th molecule belongs to the class. Therefore, we define the loss function for total training as:

$$Loss_{Total} = Loss_{CL} + Loss_C \quad (23)$$

3.4. Fusion-based prediction model

To take advantage of the idea of multi-view learning, we concatenate the 1D-SMILES, 2D-graph, and fused SMILES-graph representations to form a fusion representation. Considering the inconsistency and similarity that exists between different molecular modalities, we learn the complementary features from the corresponding network and combine all semantic and structural information to form the final fusion features which are used for the molecular ADMET properties prediction using Sigmoid function after FC layer. Notably, our algorithm strategically incorporates contrastive fusion and multi-view learning mechanisms, enabling more accurate predictions by effectively capturing the diverse features of molecular data and harmonizing their representations. The prediction pseudocode for our AMCFNet framework is shown in Algorithm 1.

4. Experiment and analysis

In this section, we will first introduce the experimental setups for datasets, evaluation, and training preparation. Then, we validate the performance of our proposed AMCFNet by ablation studies, comparative studies, and generalization studies. Therefore, all experiments are executed using GPU on the Google Colaboratory platform and Nvidia 4090, implemented with Pytorch deep learning network framework.

4.1. Experimental setups

4.1.1. Datasets

The datasets utilized in this study are sourced from the DrugBank drug molecule database at the University of Alberta and ChEMBL database (Mendez et al., 2019). ER α datasets include 1974 compounds, each associated with SMILES, descriptors, and corresponding ADMET properties. The ADMET properties represent the absorption, distribution, metabolism, excretion, and toxicity of drugs in the body. In this research, the molecular SMILES and graph are employed as the independent variables for ADMET properties prediction. The dataset is publicly available on an anonymous GitHub account (<https://github.com/dawntangcf/ER-dataset>). For comparative experiments, we evaluate our AMCFNet framework on five publicly available molecular

Algorithm 1 Pseudocode of AMCFNet prediction

Require: The drug molecules *SMILES*; The drug molecules class *Y*

Require: The drug molecules *SMILES* of number: *N*; The batch size: *B*

i $\leftarrow$ 0

Input_S, *Input_g* $\leftarrow$ { }

while *S* is not empty **do**

for all *i* from *i* to *N* **do**

Input_{Si}, *Input_{gi}* $\leftarrow$ Data_Aug(*SMILES_i*)

Input_S, *Input_g* $\leftarrow$ *Input_{Si}*, *Input_{gi}*

end for

end while

Loss $\leftarrow$ 0

while Not Converged **do**

for all each sample *i* in the batch **do**

 Compute *X_S* = K-BERT(*Input_S*)

 Compute *X_g* = Multi-GNN(*Input_g*)

 Compute *X_{S-g}* = Feedback(Fusion(MA(*X_S*, *X_g*)))

 Compute *X_{F(S-g)}* = Contrastive learning(*X_{S-g}*)

 Compute contrastive learning loss by Equation (19)

Loss $\leftarrow$ *Loss* + *Loss_{CL}*

X_F $\leftarrow$ Concatenate *X_S*, *X_g*, and *X_{F(S-g)}*

 Predict molecular ADMET properties

$\hat{Y} \leftarrow \text{Sigmoid}(\text{FC}(X_F))$

 Compute total loss by Equation (22) or (23)

Loss $\leftarrow$ *Loss* + *Loss_C*

end for

end while

Output: Predicted molecular ADMET properties $\hat{Y}$ of drug molecules

Table 1

The details of the downstream task datasets.

Dataset	Instance	Task	Evaluation
ER α _Caco2	1974	Classification	ROC-AUC (%) / ACC
ER α _CYP3A4	1974	Classification	ROC-AUC (%) / ACC
ER α _hERG	1974	Classification	ROC-AUC (%) / ACC
ER α _HOB	1974	Classification	ROC-AUC (%) / ACC
ER α _MN	1974	Classification	ROC-AUC (%) / ACC
BACE	1513	Classification	ROC-AUC (%)
BBBP	2050	Classification	ROC-AUC (%)
Clintox	1484	Classification	ROC-AUC (%)
ESOL	1128	Regression	RMSE
FreeSolv	642	Regression	RMSE

property prediction datasets, including BBBP, BACE, Clintox, ESOL, and FreeSolv. The classification tasks involve BBBP, BACE, and Clintox; while the regression tasks involve ESOL and FreeSolv. These datasets are randomly split into training set, validating set, and testing set at an 8:1:1 ratio. The specific details of the datasets are provided in Table 1.

4.1.2. Evaluation metrics

The ROC-AUC (Area under the ROC Curve) and accuracy are utilized to evaluate the performance of our AMCFNet model and baselines on ER α datasets. Meanwhile, for five comparative datasets, the ROC-AUC metric is utilized for classification tasks, and the RMSE metric is utilized for regression tasks.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad (24)$$

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (25)$$

4.1.3. Implementation details

This section presents a detailed depiction of the experimental setup. In the SMILES-based feature extraction module, the hidden unit of

Table 2
Training hyperparameters.

Parameter	Value
Learning rate	3e-5
Batch size	32
Dropout rate	0.5
Max epoch	100
Fusion dimension	128
Number of attention head	8
Optimizer	Adam
Patience	20

the pre-training model is set to 768, the attention head is set to 8, and the maximum length of the token sequence is set to 201. In the graph-based feature extraction module, the dimension of the initial input is set to 1, the dimensions of the outputs are set to 64, 64, 128, and 128, respectively, and the number of attention head is set to 8. In the adaptively contrastive fusion module, the output dimension of the FC layer is set to 128. In the subsequent downstream tasks, the batch size, learning rate, and maximum epoch are set to 32, 3e-5, and 100, respectively. In this work, we use the Adam optimizer for model training. Additionally, the dropout technique, early stop strategy, MASK strategy, and Pos_weight parameters are employed to mitigate overfitting and reduce computational costs, the validation loss is tracked each epoch, and early stopping is implemented if it fails to drop below the prior minimum for over 20 consecutive epochs. Moreover, we enhanced data by SMILES enumeration methods. For convenience, the specific parameter settings are outlined in Table 2.

4.1.4. Data augmentation in contrastive learning

Deep learning models generally depend on a significant quantity of labeled data to ensure high effectiveness and robust generalization. However, owing to the significant expenses and labor-intensive nature of laboratory experiments typically lead to a scarcity of labeled data, which increases the risk of overfitting and diminishes generalization capabilities. To this end, we conduct a SMILES enumeration strategy to enhance the diversity and quantity of molecular representations. Specifically, each compound molecule undergoes random expansion into five distinct SMILES representations based on the canonical SMILES using the RDKit library in Python. Subsequently, these SMILES representations are converted into five molecular graphs by RDKit, resulting in sets of SMILES-graph pairs for effectively obtaining complementary molecular information in the following adaptively contrastive fusion module. Some sample data are demonstrated in Table 3. However, the synonyms of molecular SMILES generated through the enumeration method hinder the model's ability to effectively distinguish between the enhanced data, resulting in decreased prediction accuracy. Nevertheless, our proposed adaptive contrastive fusion model can effectively alleviate this interference by utilizing consensus scores to contrastively learn the features of different molecular SMILES as well as the synonyms of the same molecular SMILES.

4.1.5. Model complexity analysis

The computational complexity of DL network is generally assessed by the total number of parameters and floating-point operations (FLOPs). Based on the proposed framework, we can evaluate the computational cost via the two metrics. As Table 4 shows, the number of parameters of the whole model is 208,521,665, and the number of FLOPs of the whole model is 13,835,283,418. The high number of parameters indicates that the model has a significant capacity for learning, allowing it to potentially capture complex patterns and interactions within the molecular data. Moreover, the elevated FLOPs signify the model's computational intensity. Notably, our proposed model exhibits a substantial total number of FLOPs and parameters because the input of AMCFNet requires five pairs of molecular SMILES-graph representations to learn comprehensive information, ultimately leading to improved predictive performance.

4.2. Parameter comparative experiment

To verify the optimal combination of parameters of our proposed model, we carry out comparison experiments with two parameters of adaptively contrastive fusion module. The network is configured with head set to 2, 4, and 8, while the channel size is set to 64, 128, and 256, respectively. The results as shown in the Table 5 and Fig. 2.

Here, head is the multi-attention parameter in adaptively contrastive fusion module. The value of head regulates a trade-off between the computational complexity and expressive capacity of model. As illustrated in Table 5, as the number of heads increases from 2 to 8, ROC-AUC values show a certain degree of increment in five datasets. Additionally, combining Fig. 2 with the loss/epoch curves, When the head is greater than 2, the convergence of the loss function is relatively stable. Thus, a head value of 8 is the best one for our model.

Channel represents the fusion dimension of multi-modal, aiming to achieve a balance in feature expression between 1D SMILES and 2D graph representations. As shown in Table 5, when channel is greater than 64, the ROC-AUC values have significantly increased. And in Fig. 2, when the channel is 128, the model exhibits the best loss-epoch convergence performance. So, channel = 128 is the optimal for AMCFNet model.

Based on the performance of expression capacity, loss trend, and convergence speed, we select head = 0.8 and channel = 128 for the following training network.

4.3. Ablation studies

To analyze the effectiveness of our AMCFNet framework, we perform two ablation analyses to compare the contribution of different modules and modalities with molecular ER α datasets. These consist of two parts: the molecular representations performance analysis and the adaptively contrastive fusion module performance analysis.

4.3.1. Molecular representations performance analysis

In this section, we conduct a molecular representation ablation analysis of the AMCFNet model by dissecting it into its fundamental components: Transformer-SMILES and GATs-Graph, which serve as methods for molecular feature extraction, as discussed in Sections 3.1 and 3.2. They are evaluated on five molecular properties datasets respectively. Detailed results across the various modalities are recorded in Table 6.

Transformer-SMILES means the model that predicts molecular properties using SMILES representation, while GATs-Graph refers to the model that predicts molecular properties using graph representation. AMCFNet is our proposed model, which adaptively combines both molecular SMILES and graph representations. Table 6. shows that the performance of AMCFNet overall exceeds that of its two specific-modality components. For the SMILES modality alone, the Transformer-SMILES achieves impressive scores, with ROC-AUC scores of 97.42%, 99.03%, 95.62%, 90.63%, and 99.03%, respectively. In contrast, the Graph modality displays lower performance, with ROC-AUC scores of 54.89%, 60.66%, 66.5%, 62.85%, and 69.32%, respectively, which are somewhat limited by the sparsity of the graph data. Notably, the combined use of both SMILES and Graph modalities results in the optimal performance, achieving ROC-AUC scores of 97.88% for Caco2, 99.25% for CYP3A4, 95.71% for hERG, 92.14% for HOB, and 99.48% for MN. By integrating molecular knowledge from multi-modal information, including SMILES and graphs, in our proposed AMCFNet method, more complementary molecular characteristics and accurate details can be extracted, significantly improving predictive capability and demonstrating the model's effectiveness.

Table 3
Enhanced SMILES on ER α datasets.

SMILES	Group	AUG_SMILES_1	AUG_SMILES_2	AUG_SMILES_3	AUG_SMILES_4
Oc1ccc2O[C...]	Training	C1CCC(C1)[...]	Oc1cc2c(cc1)[...]	c1c(OCCN2C...)	c1c(cc2S[C@...]
CCC(=C(c1...	Training	c1ccc(C(=C...)	C(C(c1cccc...	OC(=O)C=C...	c1c(C(=C(C...)
Oc1ccc(cc1[...]	Training	c1(ccc(C2C3(...)	c1c(ccc(c1)C1...	Oc1ccc(cc1)C...	c1(C2C3(Cc4...
CN1CCN(CC...)	Training	C1N(CCN(C1)...)	C1CN(c2ccc([...]	c12cc(O)ccc2...	C1CN(CCN1C...
CN(CCCc1cc...)	Training	C1NCCN(c2n...)	c1(O)ccc(cc1)...	C(Cc1ccc(cc...)	n1c(nc(nc1N...

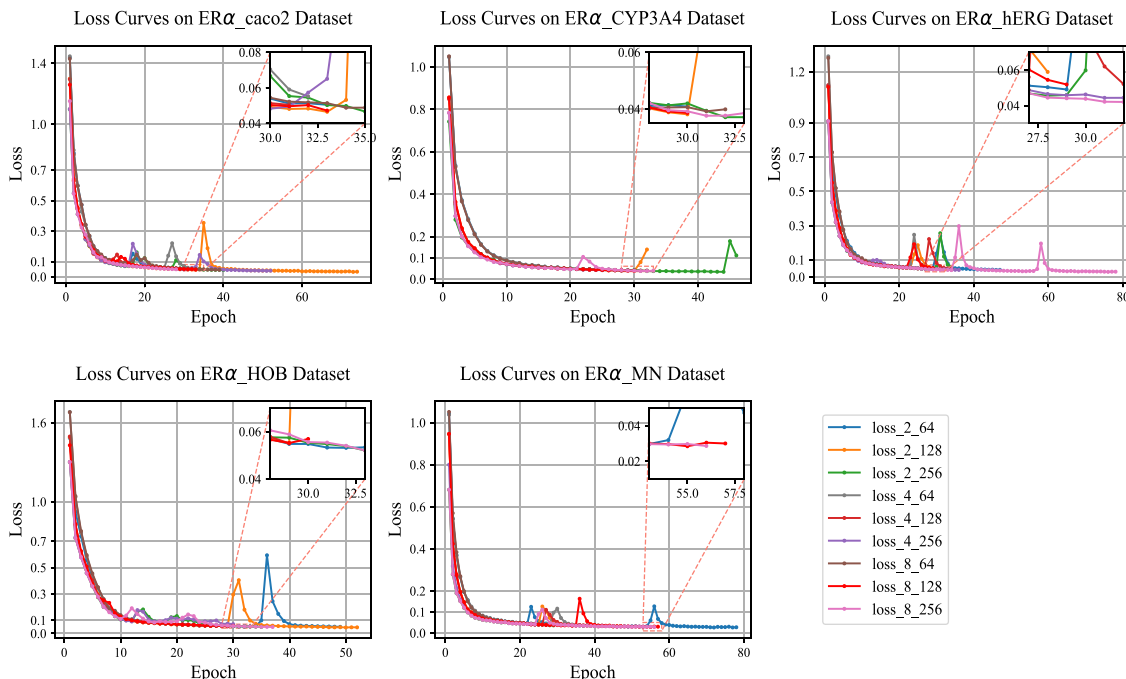


Fig. 2. Models performance on ER α datasets.

Table 4
Summary of the AMCFNet framework with parameters and FLOPs.

Module	Layer sizes	Parameters	Flops
SMILES encoder	6	199,302,912	11,678,051,760
Graph encoder	4	3,551,680	2,157,049,288
Fusion module	4	5,666,112	180,449
Prediction layer	1	961	1921
Total parameters	–	208,521,665	13,835,283,418

Table 5
Parameter comparative experiment in fusion module.

Parameter	Dataset					
Head	Channel	Caco2	CYP3A4	hERG	HOB	MN
2	64	97.38	98.37	95.01	91.70	98.69
4	64	97.43	98.51	95.72	92.25	99.05
8	64	96.74	98.25	95.85	90.70	98.04
2	128	97.60	99.17	95.46	90.17	98.97
4	128	97.48	99.02	95.43	89.10	98.87
8	128	97.88	99.25	95.79	92.14	99.48
2	256	97.12	99.14	96.48	89.74	98.68
4	256	97.39	98.47	96.39	90.64	99.43
8	256	96.97	98.37	95.62	89.76	98.85

4.3.2. Adaptively contrastive fusion module performance analysis

In this section, we evaluate the performance of our AMCFNet framework and its variants across ER α datasets. The results, as presented in the Table 7, showcase the impact of different modules on the predictive capabilities. AMCFNet-WF means model without fusion submodule, AMCFNet-WCL means model without contrastive learning submodule,

and AMCFNet-WFCL means model without both fusion and contrastive learning submodules.

The effectiveness of the fusion submodule. The fusion submodule considers interactive information between SMILES and graph representations. Model without fusion submodule demonstrates inferior performance across all molecular properties, indicating that the fusion submodule significantly contributes to predictive accuracy. This suggests that the effective integration of diverse molecular modalities is crucial, as its absence hinders the model's ability to capture complementary information from different data sources.

The effectiveness of the contrastive learning submodule. The contrastive learning submodule appears to play a role in refining the understanding of associations between molecular representations. The model without the contrastive learning submodule shows slightly lower performance in some properties compared to AMCFNet. This fact proves the potential value of the submodule, particularly in cases where it contributes to the identification of subtle relationships between molecular features.

The effectiveness of the fusion and contrastive learning submodules. There is a synergistic effect between fusion learning and contrastive learning submodule. AMCFNet-WFCL, which lacks both submodules, exhibits a decrease in performance across all properties. This highlights the joint importance of both modules, which are integral to the complete AMCFNet, enabling it to attain the best prediction accuracy. The absence of these components negatively impacts the model's ability to capture complementary features and relationships within the molecular data.

AMCFNet achieves optimal performance across four out of five data levels. The experimental results underscore the effectiveness of our proposed framework in jointly understanding diverse molecular

Table 6

Ablation studies of AMCFNet on molecular representations.

Method	SMILES	Graph	ER α _Caco2 ROC-AUC	ER α _CYP3A4 ROC-AUC	ER α _hERG ROC-AUC	ER α _HOB ROC-AUC	ER α _MN ROC-AUC
Transformer-SMILES	✓	×	97.42	99.03	95.62	90.63	99.03
GATs-Graph	×	✓	54.89	60.66	66.50	62.85	69.32
AMCFNet	✓	✓	97.88	99.25	95.71	92.14	99.48

Table 7

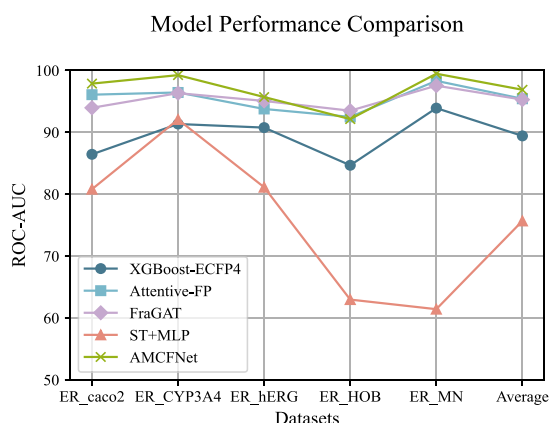
Ablation studies of AMCFNet on adaptively contrastive fusion module.

Method	Fusion	Contrastive learning	ER α _Caco2 ROC-AUC	ER α _CYP3A4 ROC-AUC	ER α _hERG ROC-AUC	ER α _HOB ROC-AUC	ER α _MN ROC-AUC
AMCFNet-WF	×	✓	97.43	99.27	94.25	91.01	98.05
AMCFNet-WCL	✓	×	96.60	99.14	95.50	90.86	98.48
AMCFNet-WFCL	×	×	97.52	99.16	95.55	91.60	91.60
AMCFNet	✓	✓	97.88	99.25	95.71	92.14	99.48

Table 8

Performance evaluation of AMCFNet with baseline methods by ROC-AUC.

Method	ER α _Caco2 ROC-AUC	ER α _CYP3A4 ROC-AUC	ER α _hERG ROC-AUC	ER α _HOB ROC-AUC	ER α _MN ROC-AUC	Average ROC-AUC
XGBoost-ECFP4	86.45	91.35	90.77	84.68	93.92	89.43
Attentive-FP	96.09	96.45	93.78	92.50	98.33	95.43
FraGAT	93.97	96.34	95.08	93.50	97.44	95.27
ST+MLP	80.81	92.08	81.13	62.97	61.42	75.68
AMCFNet	97.88	99.25	95.71	92.14	99.48	96.89

**Fig. 3.** Models performance on ER α datasets from RUC-AUC.

modalities, thereby enhancing the prediction of ER α -related molecular properties.

4.4. Comparative experiment

4.4.1. Performance of AMCFNet on ER α datasets

To compare the superiority of our proposed AMCFNet, we conduct a comprehensive evaluation of four state-of-the-art models for each ER α dataset, including XGBoost-ECFP4 (Chen and Guestin, 2016), Attentive-FP (Xiong et al., 2019), FraGAT (Zhang et al., 2021), and ST+MLP (SMILES-Transformer + Multilayer Perceptron) (Honda et al., 2019). Among them, XGBoost-ECFP4 is fingerprints-based method, Attentive-FP and FraGAT are graph-based methods, and ST+MLP is SMILES-based method. Each algorithm follows the same processing steps to maintain an equitable comparison. These methods utilize ROC-AUC and accuracy evaluation metrics to analyze the results. The specific results are shown in Tables 8, 9, and Fig. 3.

The ROC-AUC performance of our model and baseline are reported in Table 8 and Fig. 3. The results demonstrate that AMCFNet achieves optimal performance in the classification tasks of molecular ADMET properties under ER α datasets. Four of the five prediction results are

Table 9

Performance evaluation of AMCFNet with baseline methods by Accuracy.

Method	ER_Caco2 Accuracy	ER_CYP3A4 Accuracy	ER_hERG Accuracy	ER_HOB Accuracy	ER_MN Accuracy	Average Accuracy
XGBoost-ECFP4	87.85	93.16	90.89	89.11	96.20	91.44
Attentive-FP	91.88	85.79	82.71	87.82	91.37	87.91
FraGAT	87.37	91.92	89.90	87.37	95.96	90.50
ST+MLP	75.48	85.53	77.25	72.43	75.85	77.31
AMCFNet	91.88	96.45	88.83	85.28	96.45	91.78

optimal with an average ROC-AUC of 96.89%, and it also achieves the quite promising value (92.14%) on the ER α _hERG dataset. Additionally, our proposed model even reaches a ROC-AUC value of approximately 100% on the ER α _MN dataset. This can be attributed to the fact that baseline models directly learn molecular features by specific networks without considering the multi-view representations, potentially resulting in incomplete information. In contrast, the effectiveness of AMCFNet lies in its integration of molecular 1D-SMILES with 2D-graph representations, enabling it to obtain more comprehensive information than single-modal methods.

To assess our model more thoroughly, we also introduced the accuracy metric, the results are presented in Table 9. As shown, the model achieves the highest average prediction accuracy of 91.78%. In comparison to the second-best method (XGBoost-ECFP4), our proposed AMCFNet exhibits a noteworthy improvement of 0.34% in accuracy. Additionally, our proposed AMCFNet outperforms all the other methods on the ER α _Caco2 and ER α _CYP3A4 datasets. However, on the ER α _hERG and ER α _HOB datasets, it exhibits suboptimal performance. To some extent, this is due to the inherent sparsity of the graph data, which affects the model's results on specific test sets and causes discrepancies across different environments. This limitation drives us to explore more effective strategies to optimize the GNN module and reduce graph sparsity, ultimately enhancing AMCFNet's robustness and generalization.

4.4.2. Performance of AMCFNet on external datasets

To further substantiate the effectiveness and generalizability of our proposed framework for molecular property prediction, five molecular property-related publicly available datasets, including BACE, BBBP, Clintox, ESOL, and FreeSolv, are employed for this experiment, which

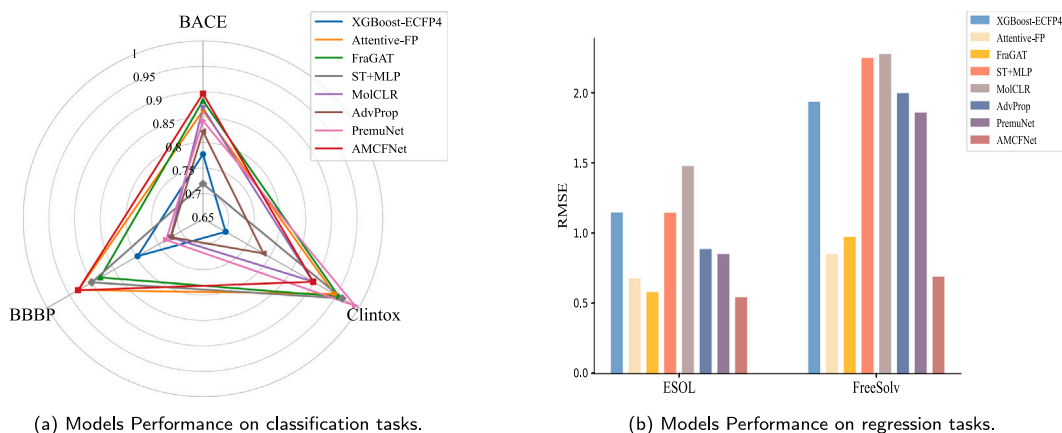


Fig. 4. Models performance on external datasets.

Table 10

Performance evaluation of AMCFNet with baseline methods on the external datasets.

Method	BACE ROC-AUC	BBBP ROC-AUC	Clintox ROC-AUC	ESOL RMSE	FreeSolv RMSE
XGBoost-ECFP4	77.70	79.70	70.10	1.146	1.934
Attentive-FP	86.30	93.01	94.54	0.679	0.850
FraGAT	88.28	88.08	95.54	0.580	0.973
ST+MLP	71.90	90.00	96.30	1.144	2.246
MolCLR	86.80	72.30	89.70	1.477	2.273
AdvProp	82.20	72.20	78.60	0.886	1.996
PremuNet	84.30	73.30	99.20	0.851	1.858
AMCFNet	89.65	93.08	89.78	0.545	0.691

covers three classification tasks and two regression tasks. It is worth noting that all methods are evaluated using the ROC-AUC metric for classification tasks and the RMSE metric for regression tasks. More detailed information is shown in Table 10 and Fig. 4. Table 10 provides an overview of the results for the external datasets, including baseline methods, while Fig. 4(a) and (b) showcase the performance of different models on classification and regression tasks, respectively. For each dataset, we utilize the current state-of-the-art graph-based algorithms (Attentive-FP (Xiong et al., 2019), FraGAT (Zhang et al., 2021), and MolCLR (Wang et al., 2022b)), fingerprints-based algorithms (XGBoost-ECFP4 (Chen and Guestrin, 2016)), SMILES-based algorithms (ST+MLP (Honda et al., 2019)), and multi-modal fusion algorithms (AdvProp (Wang et al., 2022a) and PremuNet (Zhang et al., 2024)), respectively.

Analyzing the performance across multiple tasks and methods, the results, as illustrated in Table 10 and Fig. 4, reveal the remarkable efficacy of the AMCFNet framework. When compared to the well-performing AdvProp and PremuNet multi-modal fusion models, our proposed AMCFNet framework achieves the best performance on BACE, BBBP, ESOL, and FreeSolv datasets. Specifically, our proposed model performs well in classification prediction tasks. In the BACE and BBBP datasets, AMCFNet achieves notable ROC-AUC values of 89.65% and 93.08%, respectively, demonstrating substantial improvement compared to other baseline methods. However, in the Clintox test dataset, the ROC-AUC value is 89.78%, which is somewhat limited by the class imbalance (95.75% label 0, 4.25% label 1), thus negatively impacting our model's performance. By redistributing the labels more evenly (91.97% label 0, 8.03% label 1), the ROC-AUC increased to 94.6%, matching the performance of suboptimal models. This highlights the impact of data imbalance on our model. In subsequent work, we will continue to optimize our algorithm to address class imbalance issues.

The regression task results are reported in Table 10, columns 5 and 6, and in Fig. 4(b). On the ESOL and FreeSolv datasets, AMCFNet exhibits lower RMSE values of 0.545 and 0.691, indicating superior

predictive accuracy. The AMCFNet model shows obvious performance advantages on four out of five benchmark datasets for property prediction. This is due to the observation that AdvProp and PremuNet directly integrate diverse modalities by rudimentary fusion, ignoring the interaction between the multi-view representations of different modalities. In contrast, our AMCFNet model introduces the contrastive learning strategy and similarity score function to discern associations between modalities, facilitating the adaptive fusion of multi-modal features.

These results underscore the efficacy of AMCFNet in molecular property prediction and its generalization capability, showcasing superior performance across diverse chemical tasks. The robustness of this model is evident, indicating its reliability and effectiveness in predicting molecular properties beyond the ER_a datasets of breast cancer. Moreover, this highlights the ability of AMCFNet model to learn multi-view molecular information from adaptively contrastive learning fusion structures.

5. Conclusion

In this paper, we propose an adaptively multi-modal contrastive fusion network (AMCFNet) for breast cancer that can adaptively integrate multi-modal features of molecular SMILESs and graphs. For the SMILES-based feature extraction, we introduce a prior KnowledgeBERT module to learn the molecular information of SMILESs. For the graph-based feature extraction, based on the multi-head attention, we design a multi-layer GNN to learn the molecular 2D structure of the graph representations. Additionally, an adaptively contrastive fusion module is designed to integrate the molecular SMILESs and graphs by adaptively allocating weight and contrastively learning features. Finally, we integrate the 1D-SMILES, 2D-graph, and fused SMILES-graph features for molecular ADMET properties prediction. The experimental results on the publicly available bioactivity-related datasets and state-of-the-art benchmarks demonstrate that the proposed AMCFNet effectively learns and fuses features from molecular multi-modal, which outperforms the existing models and improves the molecular ADMET properties prediction of breast cancer.

Despite the success, there remains potential for further enhancement. The proposed network is only a molecule-level fusion and learning for multi-dimensional representations, like 1D-SMILES, 2D-graph, and 3D-spatial data of a single molecule. It does not take into account other clinically significant information, such as genetic and pathological data relevant to drug discovery. And the model still has room for improvement in handling data class imbalance and graph data sparsity. In future work, we will explore a more effective and generalized multi-modal fusion method to incorporate drug molecules with pathological and genetic features to investigate potential mechanisms within drug treatment research in biological and medical contexts, while optimizing the our current model.

CRediT authorship contribution statement

Wenyan Tang: Writing – original draft, Software, Resources, Methodology, Formal analysis. **Meng Li:** Methodology, Conceptualization. **Yi Zhan:** Resources, Data curation. **Bin Chen:** Validation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the Natural Science Foundation Project of Chongqing (Grant No. cstc2020jcyj-msxmX0162) and the Chongqing Graduate Student Research Innovation Project (Grant No. yjscxx2024-284-248).

Data availability

Data will be made available on request.

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